

Social Graph-Based Restaurant Recommendation System

Leveraging Social Influence, Graph Learning, and Context Aware Ranking

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INTRODUCTION & MOTIVATION

Problem

Choosing where to eat is difficult because most recommendation systems rely on global ratings, which often don't reflect what a user prefers. They also make limited use of social connections, even though people tend to trust recommendations from others with similar tastes.

We address this by modeling restaurant recommendation as a personalized ranking problem on a social graph, combining user preferences, social influence, and contextual signals.

Importance

Our approach works better than traditional methods because restaurant discovery is naturally influenced by social connections and context

DATA

We use the Yelp Open Dataset (Official Academic Release) which includes:

- 6.75M reviews, 1.99M users, and 35K restaurants
- Around 9GB of data in JSON format
- Data collected from 2010 to 2024

Data Processing

We combine ratings, social connections, and business information to capture richer recommendation signals.

- Multimodal signals: ratings, social ties, and business attributes
- Data filtering to keep relevant and reliable interactions

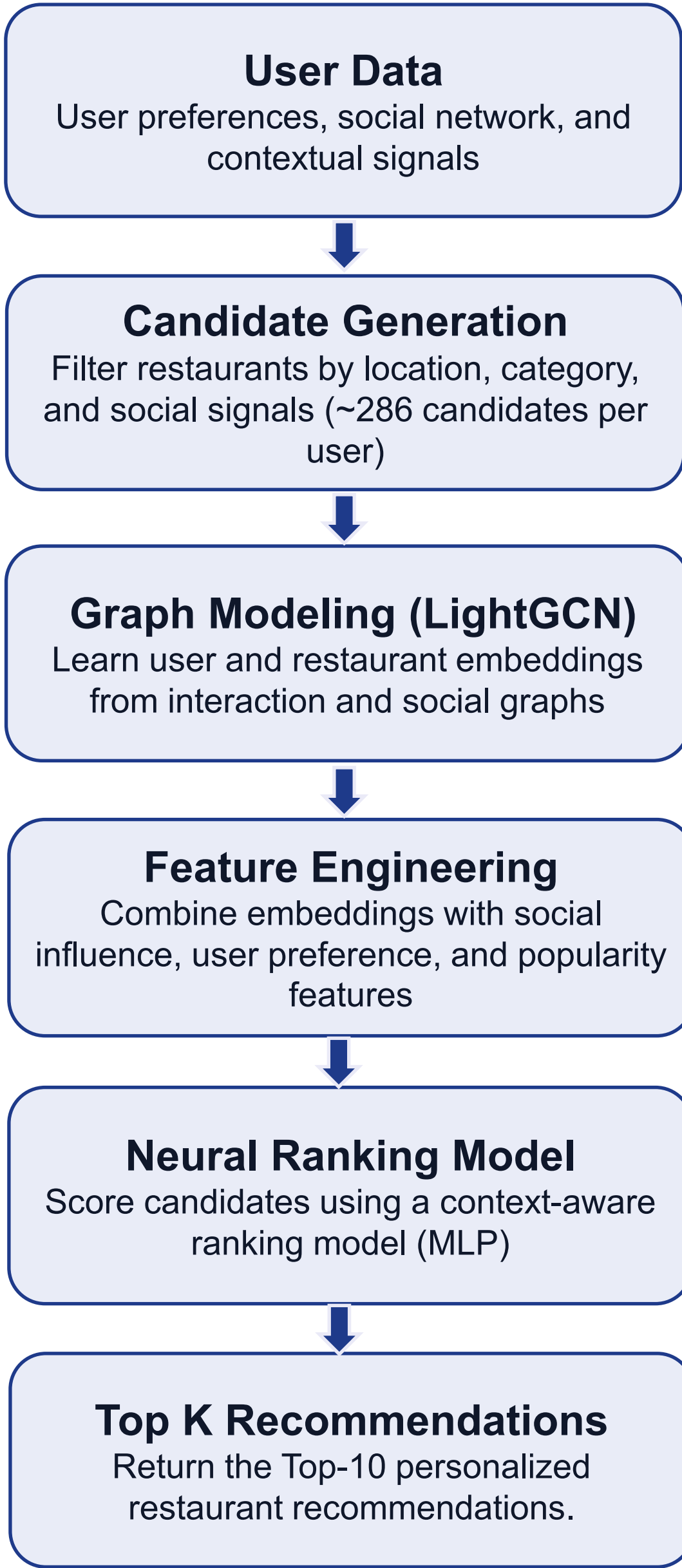
Model Design Comparison

This table summarizes the key design differences between our approach and standard recommendation models.

Feature	MF	NGCF	LightGCN Hybrid (Ours)
Relational Depth	1-Hop	Multi-Hop	Multi-Hop
Social Logic	None	Complex	Regularized (Jaccard)
Spatial Awareness	None	None	Pre-Screening
Edge Weighting	Uniform	Uniform	Jaccard-weighted
Scoring	Single	Embedding	Transparent (Hybrid)

APPROACH

Our system follows a two-stage pipeline that separates candidate retrieval from ranking. This improves efficiency by reducing the search space before applying a more expressive model.



LightGCN Mechanism

Aggregates information from similar users in the interaction graph to learn user preferences:

$$e_u^{(k+1)} = \sum (i \in N_u) 1/\sqrt{(|N_u||N_i|)} e_i^k$$

Jaccard Social Weighting

Measures similarity between users based on shared restaurant interactions:

$$J(u_1, u_2) = \frac{|I_{u1} \cap I_{u2}|}{|I_{u1} \cup I_{u2}|}$$

Neural Ranking (MLP)

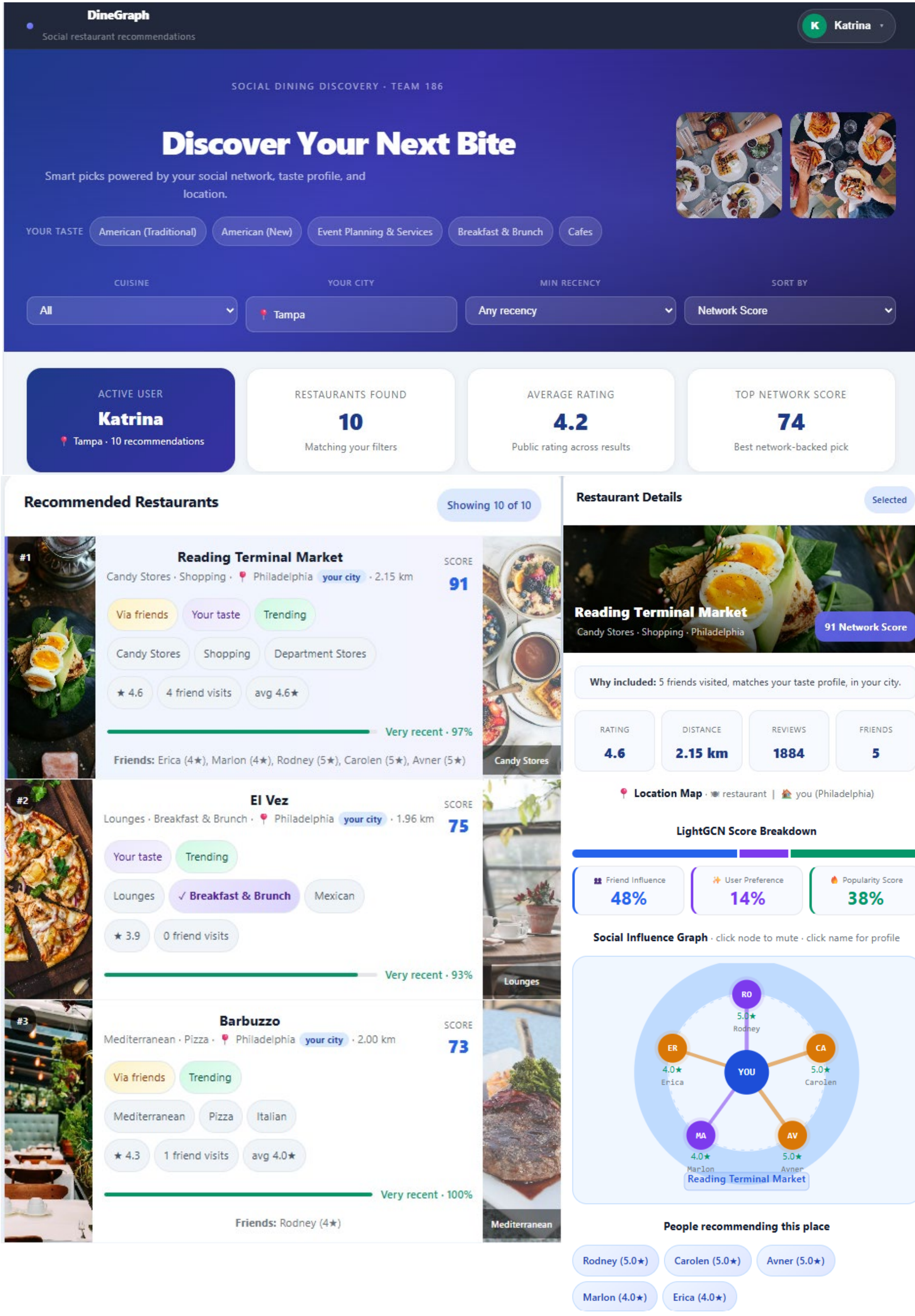
A neural model combines learned representations and contextual features to rank candidate restaurants.

UI VISUALIZATION

The system provides an interactive interface for exploring personalized restaurant recommendations.

Key Features

- Interactive social graph** showing how friends influence each recommendation
- Score breakdown** that explains recommendations using social, preference, and popularity signals
- Source-aware labels** (e.g., friends, personal taste, trending) to highlight why items are recommended
- Dynamic interaction** that allows users to explore and compare recommendations



KEY INNOVATIONS

1

Efficient Candidate Generation



Context-aware two-stage retrieval reduces the search space from ~35K to ~286 candidates per user using geographic, category, and social filtering. This improves efficiency while ensuring that only relevant and accessible options are considered, reducing decision fatigue and enabling scalable recommendation.

2

Jaccard Weighted Social Influence



Socially-Regularized Weighting is based on shared restaurant interactions, capturing behavioral similarity between users. This prioritizes recommendations from like minded users while filtering out irrelevant social noise, improving personalization and trust in recommendations.

3

Transparent Hybrid Scoring



Explainable multi-signal ranking combines graph embeddings (LightGCN) with features like user preference, popularity, and context. This helps capture more realistic user preferences while making it clear why a restaurant is recommended through a simple score breakdown.

EXPERIMENTS AND RESULTS

Experimental Setup

We evaluate on the Yelp Open Dataset using a temporal split to mimic real-world recommendations. The dataset includes **17,221 active users** and **35,000 restaurants**, with **435,744 interactions** held out for testing.

Performance is measured using Precision@10, Recall@10, HR@10, and NDCG@10. To improve efficiency, we restrict ranking to a candidate set of **about 286 restaurants per user**.

Performance Metrics Comparison

Model	Prec@10	Recall@10	HR@10	NDCG@10
LightGCN Hybrid (Ours)	0.0426	0.0783	0.3445	0.1832
Popularity Baseline	0.0377	0.0675	0.3066	0.1636
Category + Popularity	0.0166	0.0318	0.1509	0.0789

Key Results

- Our model improves **Precision@10** by about **13%** compared to the popularity baseline
- We see a **~12% increase in NDCG@10**, and it means more relevant restaurants are ranked higher in the list.
- The system **reduces the search space from ~35K restaurants to ~286 candidates per user**, while maintaining strong performance.

Improvement Over Baseline

+13% Precision@10	+16% Recall@10	+12% NDCG@10
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Why It Works

These improvements come from combining **social and preference signals**, which better capture personalized "word-of-mouth" behavior than global popularity alone. LightGCN further improves performance by learning higher order relationships between users, while the two-stage pipeline keeps the system efficient without sacrificing relevant options.